

Chemistry Ch 11, Part 1: Atmosphere aur Air Pollutants

Aasan Notes for Seemab · kitab ke page 153 se 156 tak · 11.1 aur 11.2.1 se 11.2.6

Pehle ek sawal. Sardiyon mein har saal Pakistan mein kuch log gas heater chalu chhor kar so jate hain aur subah nahin uthte. Ab socho: gas leak ho to bu aati hai, insan jaag jata hai. To phir ye log jagte kyun nahin? Jawab is note mein hai, aur ek dafa samajh lo to zindagi bhar yaad rahega.

1. Atmosphere kya hai (p155)

Kitab ki definition

"The envelope of gases and water vapour surrounding the planet Earth is called the atmosphere."

Yani zameen ke gird gases aur water vapour ka **ghilaf**.

Hawa mein kya kya hai

Dry air: **78% nitrogen, 21% oxygen**, baqi noble gases aur carbon dioxide (Figure 11.1).

Water vapour alag hai aur wo fix nahin hota: registan ki hawa mein bohat kam, aur **tropical rainforest mein 4% tak**. Kitab kehti hai ye "varies from place to place and from time to time". Imtihan mein ye 4% wala number aksar poochha jata hai.

2. Pollutant kise kehte hain (p155)

Kitab ki definition, lafz ba lafz

"Anything that is in the air, water or soil which has a harmful effect on some part of the environment is called pollutant."

Kitab mein ek ghalti chhapi hai

Is section ka number kitab mein ~~10.2 AIR POLLUTANTS~~ likha hua hai. Ye ghalat hai, hona chahiye tha **11.2 AIR POLLUTANTS**, kyunke ye Unit 11 hai. Aage ke saaray sub-sections theek se 11.2.1, 11.2.2 likhe hue hain. **Imtihan mein 11.2 likhein.**

3. Chhe air pollutants (p155, p156, Figure 11.2)

Pollutant	Kahan se aata hai	Kya nuqsan karta hai
Sulphur oxides SO_2, SO_3	Sulphur fossil fuels mein qudrati taur par hota hai. Power plants, vehicles, industrial units, generators aur gharon ke heating systems.	Powerful irritant ; respiratory system mein jazb ho jata hai; asthma, bronchitis, emphysema ko barha deta hai. Acid rain aur haze ka zimmedar.
Nitrogen oxides NO aur $\text{NO}_2 = \text{NO}_x$	Sab se bara source: vehicles, power plants, industrial units aur generators mein fossil fuels ka jalna . Car engines bohat zyada banate hain.	Bohat zehreele ; acid rain, photochemical smog , sar dard aur saans ke amraz.
Carbon dioxide CO_2	Qudrati taur par bhi hai, lekin insani kaam se bohat barh gaya: fossil fuels jalana, deforestation , industrial processes.	Climate change aur global warming .

Carbon monoxide CO	Incomplete combustion , yani jab jalne ke liye oxygen पूरी na ho. Vehicles, industrial processes, aur lakri ya gas se chalne wale heating systems.	Zehreela . Hemoglobin ke saath mil kar carboxyhemoglobin banata hai, jo oxygen le hi nahin ja sakti. Behoshi, dam ghutna, aur maut.
Methane CH ₄	Sabzi aur kachray ka galna; wetlands, rice fields, landfills ; aur livestock , yani gaay, bakri, bhairoh ka nizam e hazm.	Global warming barhata hai, jo climate change ki wajah banta hai.
Ground level ozone (smog)	Ye seedha nahin nikalta . Nitrogen oxides aur VOC (volatile organic compounds) ke darmiyan sunlight mein reaction se banta hai.	Shehri aur industrial ilaqon ka smog.

Ek lafz jo imtihan mein number deta hai: secondary pollutant

Baqi paanch pollutants **seedha** chimney ya exhaust se nikalte hain. Lekin **ground level ozone kisi chimney se nahin nikalta**: wo hawa hi mein, dhoop ki madad se, do doosray pollutants se ban jata hai. Isi liye kitab ise **secondary pollutant** kehti hai. Ye lafz zaroor likhna.

Ab shuru wala sawal

Gas heater wale log isi liye nahin jagte kyunke jo gas unhein maar rahi hoti hai wo **carbon monoxide** hai, aur kitab us ke bare mein saaf likhti hai: "Carbon monoxide is a colourless and odourless gas, so its presence cannot be felt."

Na rang, na bu. Insan ko pata hi nahin chalta. Aur andar ja kar wo khoon ke hemoglobin ke saath chipak jati hai aur **carboxyhemoglobin** bana deti hai, jo oxygen utha hi nahin sakti. Yani saans to chal rahi hai, hawa bhi andar ja rahi hai, magar oxygen jism tak pohanch nahin rahi. Insan behosh hota hai aur phir dam ghut jata hai.

Aur ye kab banti hai

Jab **oxygen पूरी na ho**, yani **incomplete combustion**. Isi liye band kamray mein heater khatarnak hai: oxygen khatam hoti jati hai, aur jitni oxygen kam, utni zyada CO. **Kamray mein hawa ka guzar zaroori hai**. Ye sirf imtihan ka sawal nahin, ye ghar ki baat hai.

Quiz

Ye teeno sawal tumhare Chemistry paper mein bhi aayenge, isi liye inhein achhi tarah karo.

1. Define **atmosphere** and **pollutant** in the words of your book. Then state the percentage of nitrogen and of oxygen in dry air, and the maximum percentage of water vapour the book gives for a tropical rainforest.
2. Name the **six** air pollutants in Figure 11.2. For **sulphur oxides** and for **nitrogen oxides**, write the chemical formulas and one source and two harmful effects each.
3. (i) How is carbon monoxide produced, and why can a person not detect it? (ii) Explain what it does inside the body, naming the compound it forms with haemoglobin. (iii) Why is ground level ozone called a **secondary pollutant**?